# Concept

## Diagram Recipes

When things relate to other things.

IN THIS CAROUSEL

# What you'll learn.

A recipe for the concept diagram: when to reach for it, how to build it, and what to watch for.

**01** **What & Why** — **3** SLIDES

What a concept diagram is and when it fits.

**02** **How to Build** — **3** SLIDES

Three steps: shape, connect, and label.

**03** **What to Expect** — **2** SLIDES

Ingredients, ideas, and the key watch-out.

# Concepts that relate. Draw them as circles.

When concepts, objects, or systems relate to each other but do not nest inside each other, concept diagrams show how they connect and what those connections mean.

**DO**

Use when things relate, associate, influence, or depend on each other: systems, teams, components, workflows.

**DON'T**

Use for containment or nesting: that needs a block diagram. Use for timelines or sequences: those need different diagram types.

ALSO KNOWN AS

ERD · mind map · bubble chart


pk@getpraxium.ai

# Every system has these relationships.

The connections already exist. A concept diagram names them and shows what kind they are.

IN A VEHICLE

**BMS ⟷ Thermal Controller ⟶ Pump**

BMS and Thermal Controller exchange data both ways. The controller drives the pump one way only. The line type tells you who controls whom.

IN A CHIP TEAM

**RTL Design ⟷ Verification ⟶ Sign-off**

Design and verification iterate together. Verification results drive sign-off in one direction only. Same roles, very different relationships.

IN A PRODUCT LINE

**Platform —— Tier 1 Supplier —— OEM**

A platform associates with its supplier and its OEM customer. Neither controls the other. The undirected line is a deliberate choice, not a default.

# 3 steps to build it.

Start with shapes, then connect them, then label each connection.

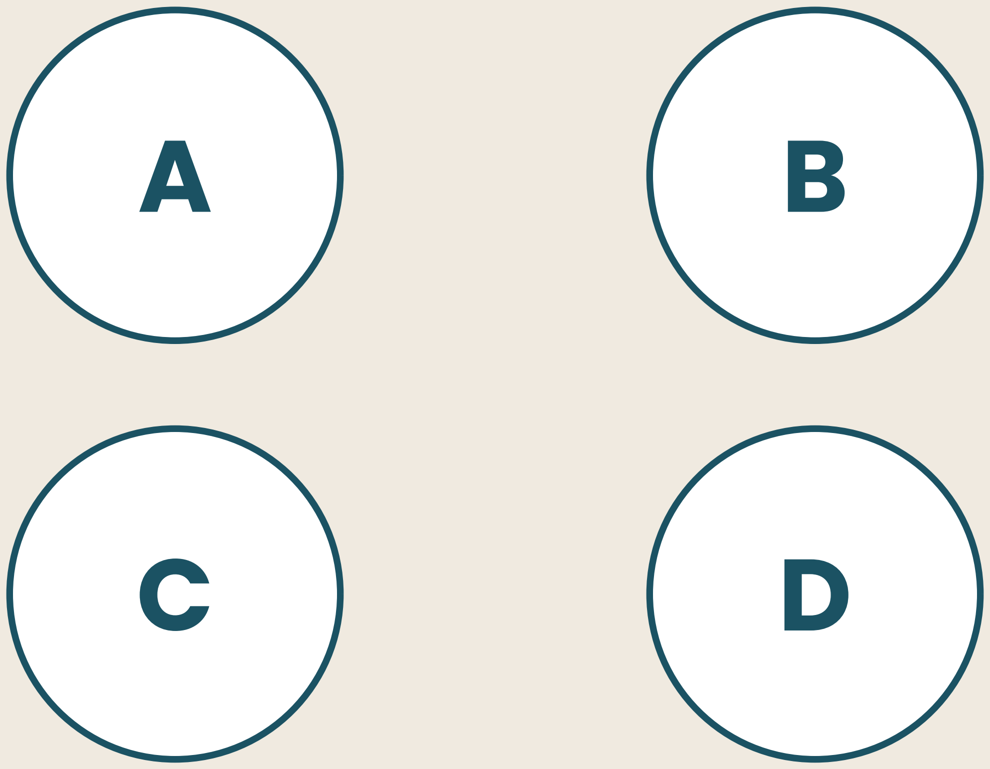
**01** **Shape**
One circle per concept, object, or system in scope. No merging, no splitting.

**02** **Connect**
Join circles with a line. Choose the right line type for each relationship.

**03** **Label**
Write the relationship on each line so it reads as a complete statement.

# Step 1 — Shape. One circle per concept.

Every concept, object, or system in scope gets its own circle. One idea per shape. Do not merge two concepts into one circle.




FOR EXAMPLE

**BMS · Thermal Controller · Coolant Pump · Battery Pack**

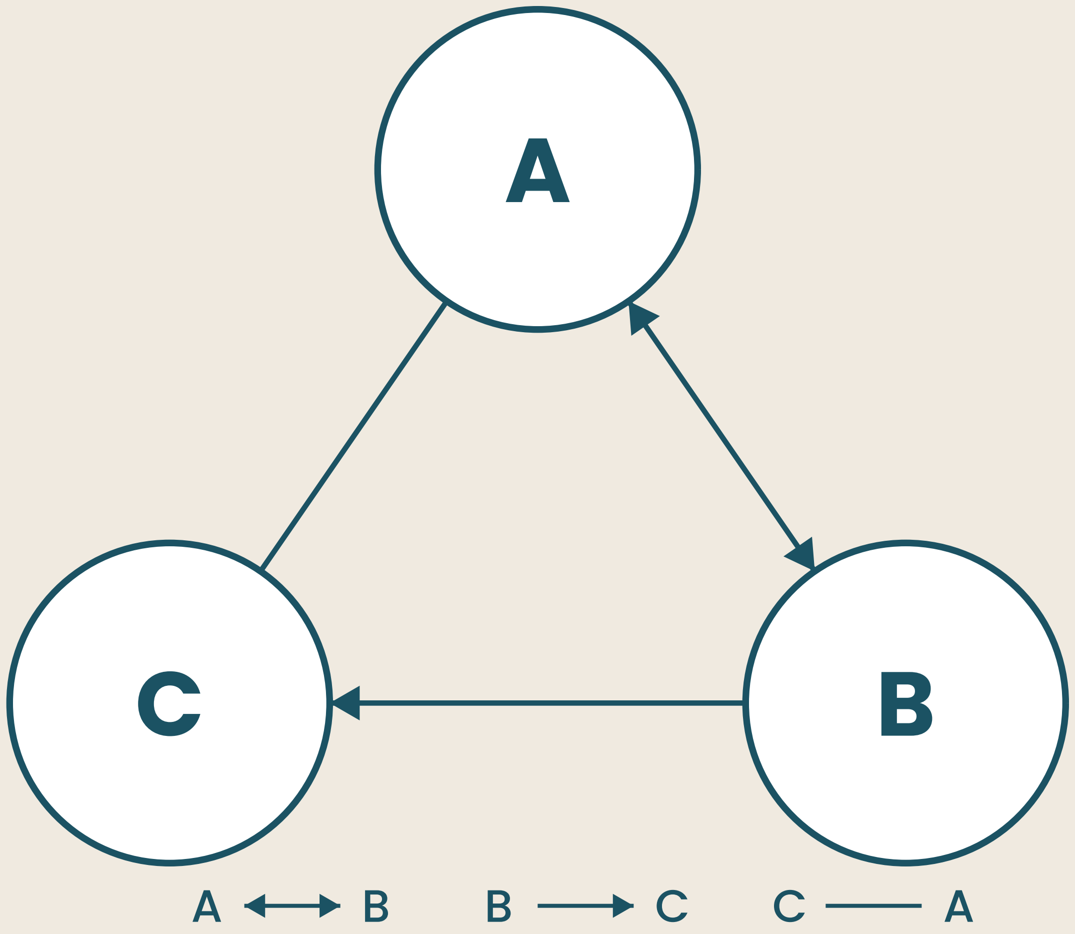
# Step 2 — Connect. Choose the right line.

Three line types. Each encodes a different kind of relationship. The choice is not decoration. It is the information.

**Non-directional**
for associations

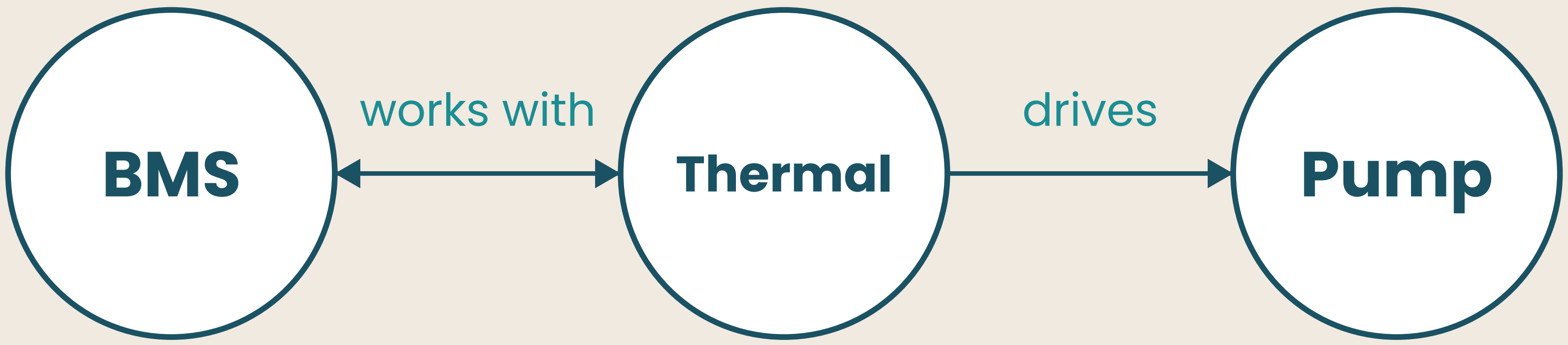
**One-way**
for sequences

**Two-way**
for reciprocation



# Step 3 — Label.
# Make lines readable.

Write the relationship on each line. Aim for a phrase that completes the sentence: "A [label] B."




"BMS works with Thermal Controller. Thermal Controller drives Pump." The diagram reads as statements.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# Let line type carry the meaning.

Three line types exist to encode direction. Use the right one for each relationship, every time.

**DO**

Vary line types to reflect the real direction of each relationship. A non-directional line is a deliberate choice, not a default.

**DON'T**

Use the same line type for every connection. A diagram of identical lines makes all relationships look equivalent and forces the reader to guess.

FROM THE SOURCE

**Set line type after you understand the relationship, not before.**

# Two ingredients. One diagram.

Concept diagrams need less to start than most. Gather these two things before you draw.

**01** **A list of every concept, object, or system in scope.**

Be strict about scope. Every node added outside scope makes the diagram harder to read.

**02** **An understanding of how each pair of things relates.**

Know the direction before you draw the line. Association, sequence, reciprocation: each is a different diagram.

# Three places to start.

Concept diagrams work wherever things relate, depend on, or influence each other.

**01** **System interfaces in hardware design**

Show how ECUs, sensors, and actuators relate and which connections are one-way vs. shared.

**02** **Team and stakeholder relationships**

Map who works with whom, who reports to whom, and who influences whom across your organisation.

**03** **Product and supplier networks**

Show how platforms, suppliers, and customers associate without forcing a hierarchy that does not exist.

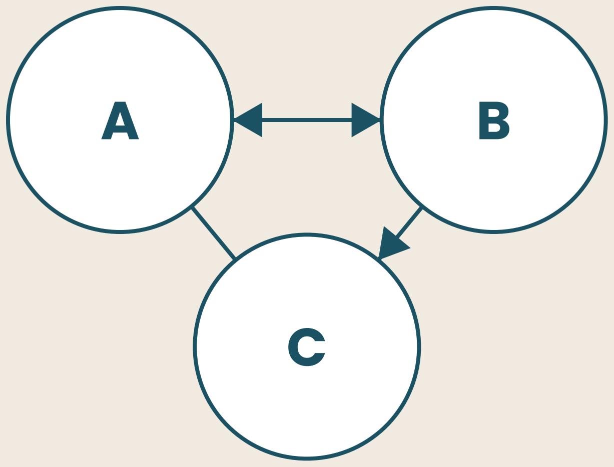
# Concept.

## STUCK? Diagrams Help.

This recipe is from STUCK? Diagrams Help., a guide to 12 diagrams every professional should know. See page 222.

ALSO KNOWN AS

ERD · mind map · bubble chart

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